

Chapter 10

Lighting

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Learning Outcomes

After studying this chapter, you are expected to be able to

1. Briefly describe light and colour interaction, and how surface material absorbs and reflects light, and how the eye perceives colour.
2. Explain how the colour of a fragment can be calculated using illumination, the colour of a light source, and the colour of a surface material.
3. Explain how Phong's lighting model can be used to calculate the illumination of a point on a surface.
4. Explain how to calculate the three components of Phong's lighting model (ambient, diffuse and specular), and how they relate to the physical world.
5. Recognise the repeated use of dot product in lighting calculations.
6. Explain the difference between the three well known standard light types in computer graphics, (directional light, positional light, and spot light).
7. Compare and distinguish the difference in the effect on the illumination of a fragment by the three well known standard light types in computer graphics (directional light, positional light, and spot light), and decide which light type you want to use.
8. Explain how to use Phong's lighting model to implement the three well known standard light types in computer graphics (directional light, positional light, and spot light).
9. Illustrate the illumination of a fragment by the three well known standard light types in computer graphics (directional light, positional light, and spot light).
10. Define the three well known standard light types in computer graphics (directional light, positional light, and spot light) parametrically.

Colour Perception

In the physical world, the colour of a surface that the eye perceives depends on the light which is reflected off the surface and reaches the nerves at the back of the eye.

Light is made up of photons. The eye perceives colour by nerve cells at the back of the eye becoming excited by photons with different wave lengths. There are 3 types of nerve cells in the eye, some which excited by red light which has a wavelength of 620–750 nm, some which get excited by green light which has a wavelength of 495–570 nm, and some which get excited by blue light which has a wavelength of 450–495 nm.



Image from science.nasa

We can perceive all different colours by mixtures of these primary colours.

Light sources emit light of a given colour according to the wavelengths of the photons. White light is made up of equal amount of all wavelengths.

Surface materials absorb or reflect light depending on the colour of the light and the colour of the material.

The reflected light is calculated using vector component multiplication (we will use this $\otimes$ symbol).

For example, assuming a white light source and a red material, the reflected light is a red light. This is why red objects appear red in white light. On the other hand, assuming a white light source and a black material, the reflected light colour is a black colour. This is why black objects appear black in a white light (or in fact any coloured light).

Light colour	Surface Colour	Reflected Colour
$s = (1.0, 1.0, 1.0)$	$m = (1.0, 0.0, 0.0)$	$r = (1.0 * 1.0, 1.0 * 0.0, 1.0 * 0.0) = (1.0, 0.0, 0.0)$
$s = (1.0, 1.0, 1.0)$	$m = (0.0, 0.0, 0.0)$	$r = (1.0 * 0.0, 1.0 * 0.0, 1.0 * 0.0) = (0.0, 0.0, 0.0)$

The colour of a fragment is calculated as the combined surface and light source colours (the colour not absorbed by the surface material) scaled by the total illumination of a triangle's fragment

$$fragment_colour = i \cdot (light_colour \otimes surface_colour)$$

Phong lighting

A *local lighting* model considers the light from light sources only rather than considering light bouncing off other surfaces. *Phong lighting* is a local lighting model that was formulated by Bui Tuong Phong in 1975 (Phong, 1975).

The model specifies that the total illumination of a triangle's fragment is the sum of ambient, diffuse and specular components.

$$i_{phong} = i_{amb} + i_{diff} + i_{spec}$$

Phong's model can be used to colour the fragment

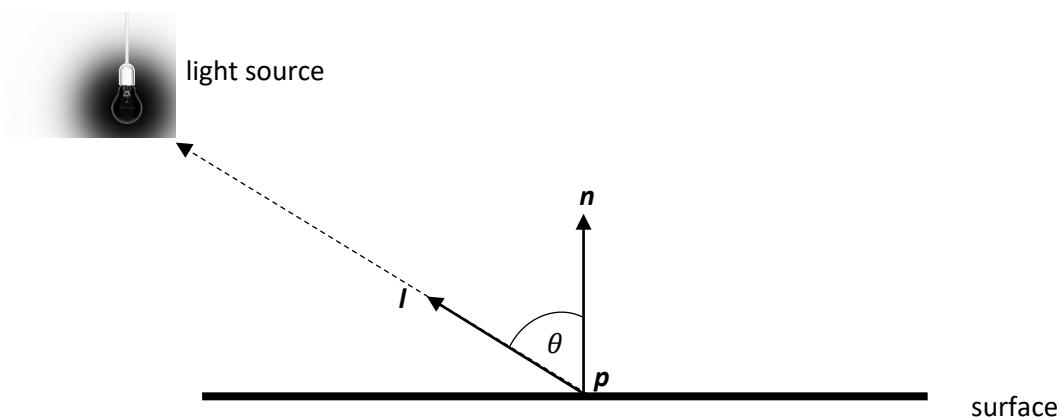
$$fragment_colour = i_{phong} \cdot (light_colour \otimes surface_colour) \quad (1)$$

The ambient component i_{amb} accounts for some light reaching surfaces which face away from a light source, which can be a result of light bouncing off other surfaces before reaching the one that is being calculated. It is often approximated using a constant k

$$i_{amb} = k$$

The diffuse component i_{diff} is based on Lambert's Law (Lambert, 1760) which states that the reflected light at a point $\mathbf{p}$ on a surface is equal to the cosine of the angle between the surface normal $\mathbf{n}$ and the vector $\mathbf{l}$ from $\mathbf{p}$ to the light source (both $\mathbf{n}$ and $\mathbf{l}$ are normalized vectors with unit length).

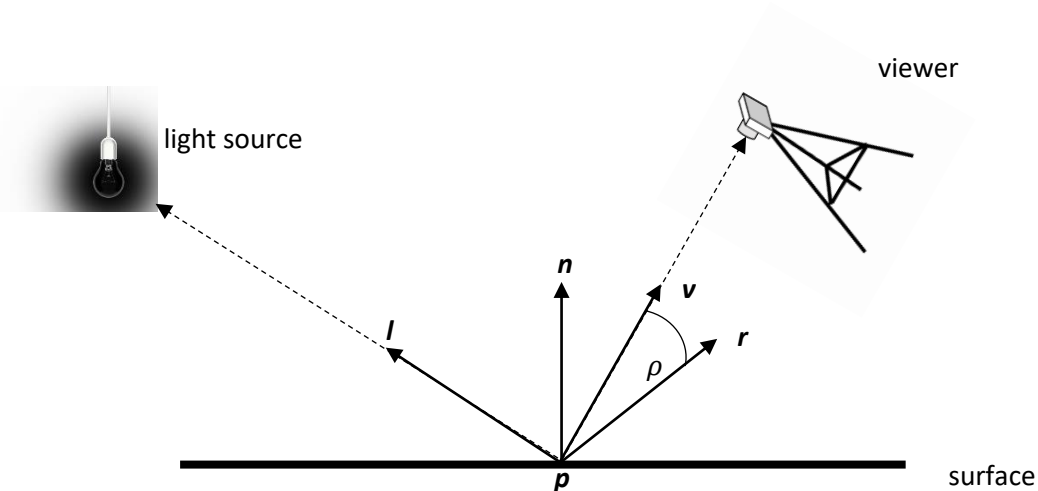
$$i_{diff} = \mathbf{n} \cdot \mathbf{l} = \cos \theta$$



The viewer's location is not a part of the calculation for the diffuse component.

The specular component i_{spec} represents highlights which give indication of the location of a light source. It states that reflected light at a point $\mathbf{p}$ on a surface is equal to the cosine of the angle between the vector $\mathbf{r}$ which is the reflected light around the surface normal $\mathbf{n}$, and the vector $\mathbf{v}$ from $\mathbf{p}$ to the viewer and the (both $\mathbf{r}$ and $\mathbf{v}$ are normalized vectors with unit length)

$$i_{spec} = \mathbf{r} \cdot \mathbf{v} = \cos \rho$$



The viewer's location is a part of the calculation for specular. The reflected light scatters most strongly towards the viewer.

The specular component is further modified by an additional parameter which is called shininess, shi

$$i_{spec} = (\mathbf{r} \cdot \mathbf{v})^{shi} = (\cos \rho)^{shi}$$

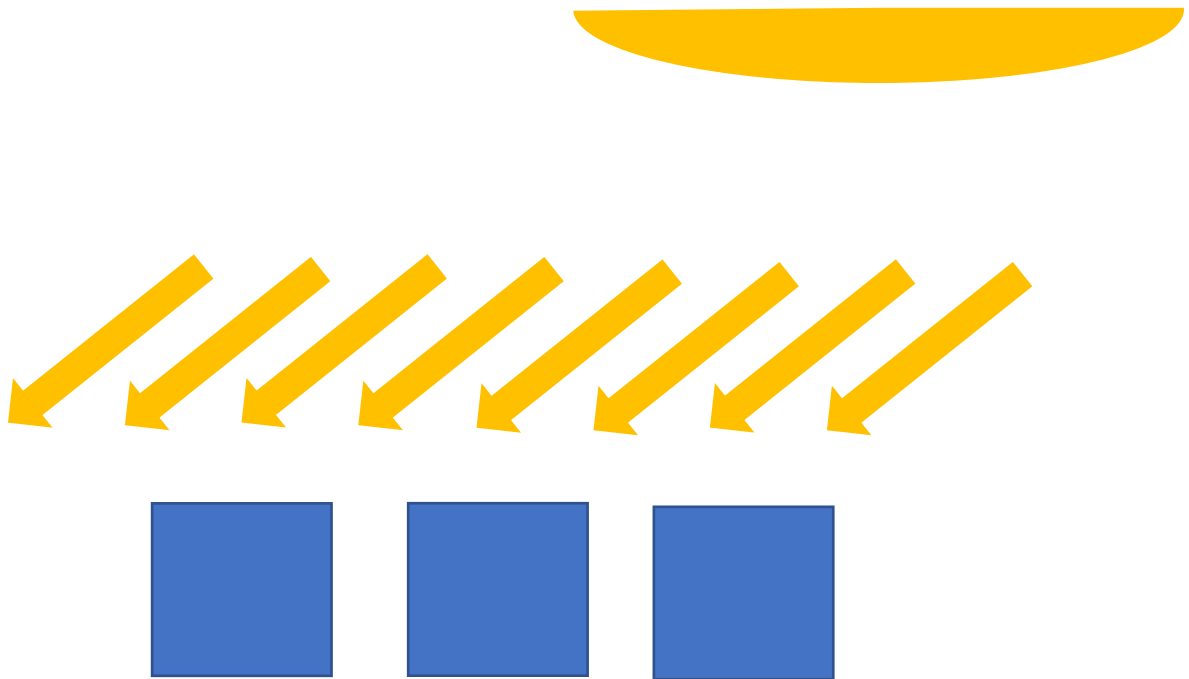
shi tends to have values $1 \leq shi \leq 256$. The larger the value of shi the faster the specular intensity decreases to 0 as the angle between the reflected light $\mathbf{r}$ and viewer $\mathbf{v}$ increases.

The specular component is not based on physics like diffuse is, however it looks fine.

Directional Light

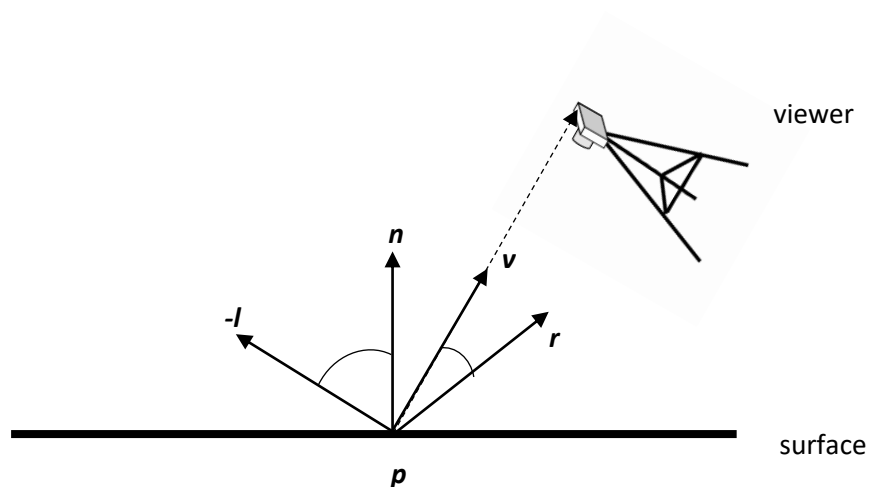
A *directional light* is considered to be infinitely away from the scene. The sun can be thought of a directional light because it is so far away that its effect appears to just have a direction.

Since the light source does not have a position, then all fragments are lit with the same illumination regardless of their position.



A directional light is defined by

1. a colour vector
2. a direction vector, $l = (l_0, l_1, l_2)$, towards which the light is shining



Phong's model can be used to calculate the total illumination of a fragment by a directional light used in equation (1) where Phong's model is calculated as

$$i_{phong} = i_{amb} + i_{diff} + i_{spec}$$

The ambient component is set to a constant k

$$i_{amb} = k$$

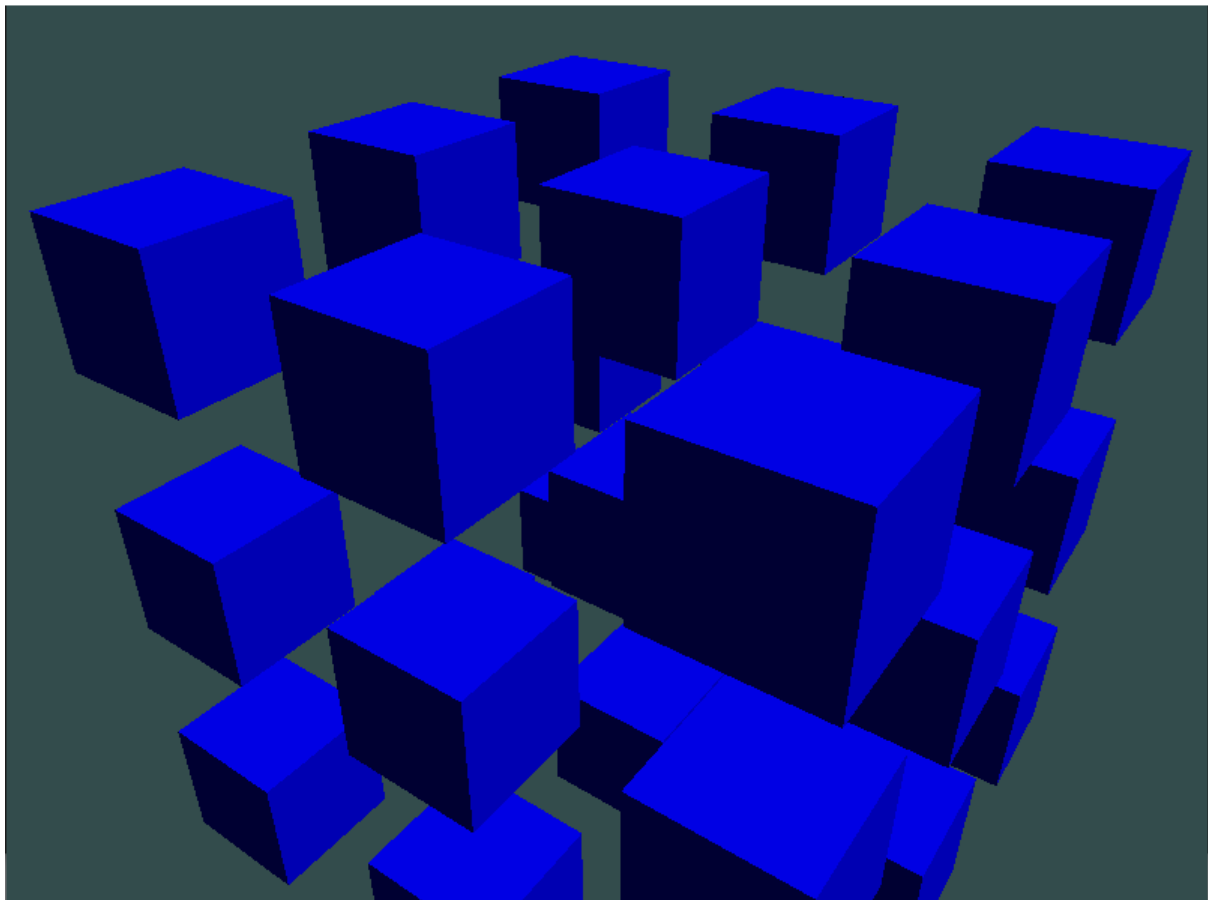
The diffuse component is the dot product of the normal of the fragment with the light direction from the definition of the light

$$i_{diff} = \max(n \cdot -l, 0)$$

The specular component is calculated as the dot product of the reflected light and the vector to the viewer, raised to the power of shi

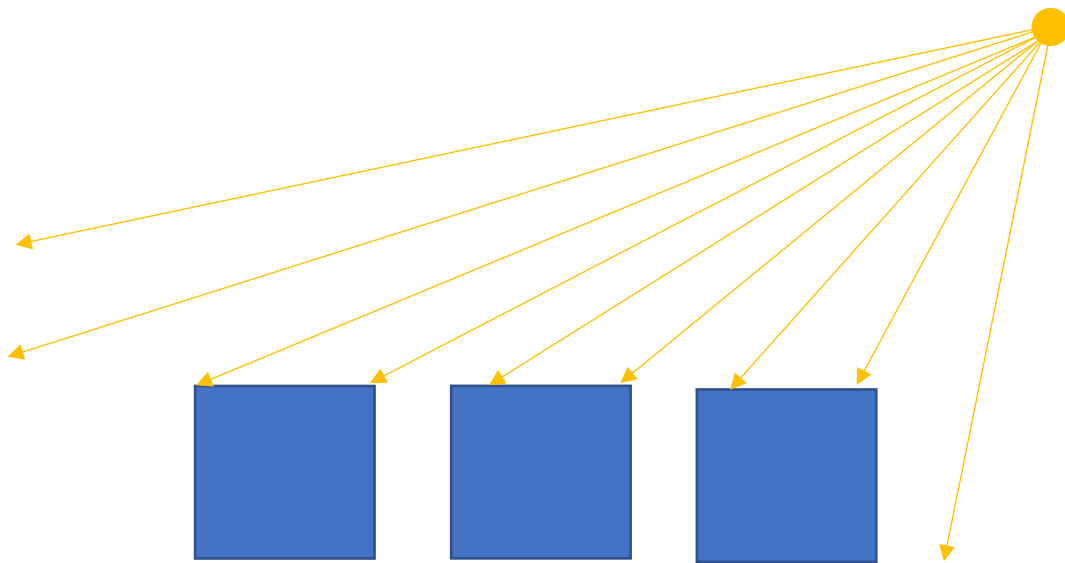
$$i_{spec} = \max(r \cdot v, 0)^{shi}$$

In the image below, surfaces of rendered cubes can be seen and are lit by a directional light. The top face of every cube is illuminated by the same amount because the light does not have a position.



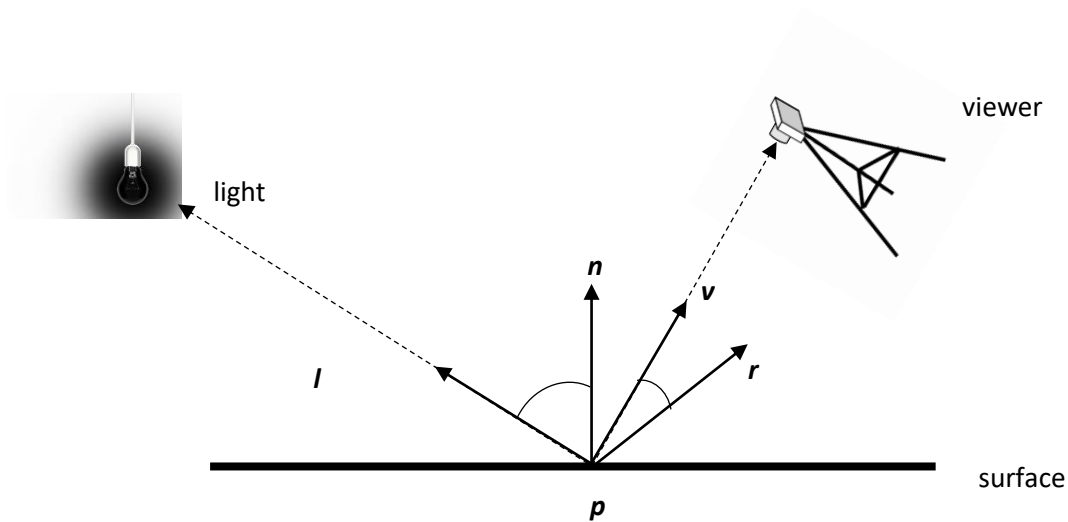
Positional Light

A *positional light* is considered to have a position and therefore a measurable distance away from each of the fragments in the scene. A light attached to the ceiling in the room you are sitting can be thought of as a positional light, and the effect of the light on different surfaces relates to their positions.



A positional light is defined by

1. a colour
2. a position, $light_pos$
3. constant attenuation, att_c
4. linear attenuation, att_l
5. quadratic attenuation, att_q



Phong's model can be used to calculate the total illumination of a fragment by a positional light in more or less exactly the same way as it was used for calculating the illumination by a directional light.

Phong's model is now calculated as

$$i_{phong} = (i_{amb} + i_{diff} + i_{spec}) * attenuation$$

Since a positional light has a position, we can use the attenuation value to scale the illumination. The illumination of a fragment based on its distance to a light source is called attenuation, where

$$attenuation = \frac{1}{att_c + (att_l * d) + (att_q * d^2)}$$

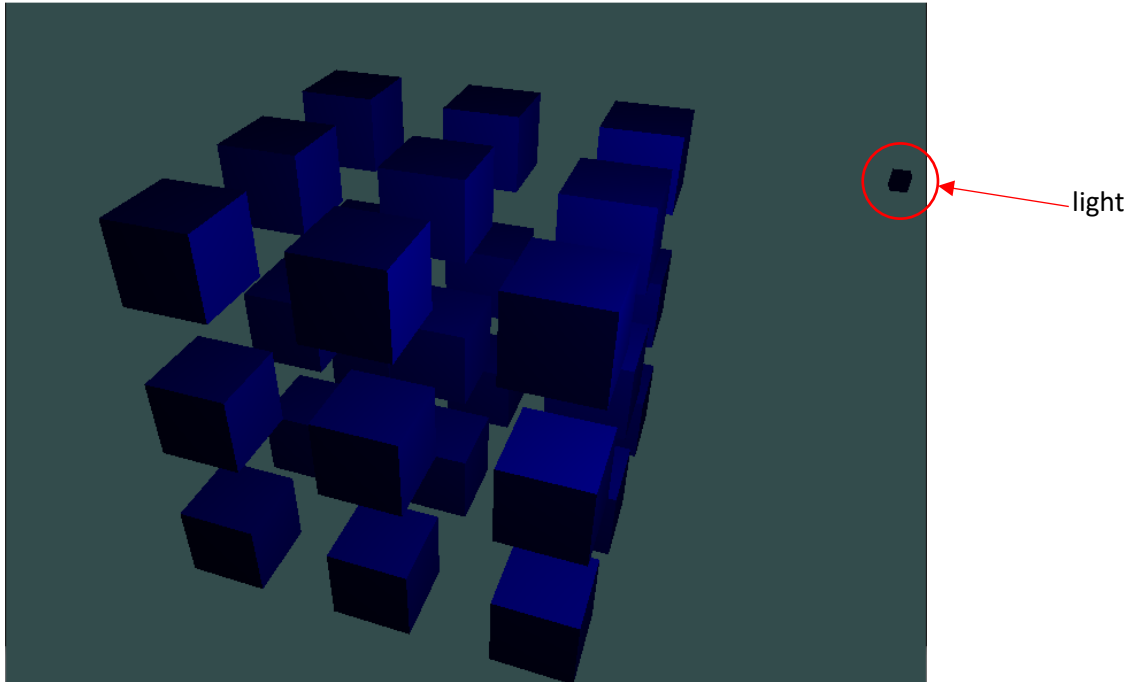
where d is the distance between the fragment and the light source. Since the light source has a position, then fragments are lit with varying amount of illumination based on the distance from the light source.

Since a positional light has a position instead of a direction, the light direction needs to be calculated using the fragment position and light position

$$l = normalise(light_pos - p)$$

$$i_{diff} = \max(n \cdot l, 0)$$

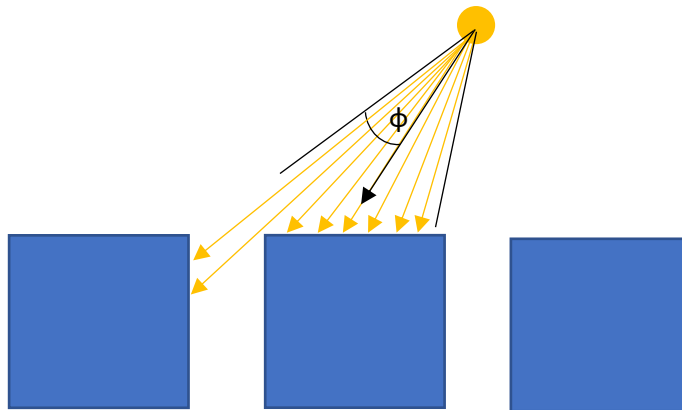
In the image below, we can see that the surfaces which are closer to the light are illuminated more than those surfaces further from the light. The light itself is rendered as an unlit cube on the far right of the image, and is indicated with a red circle.



Spot Light

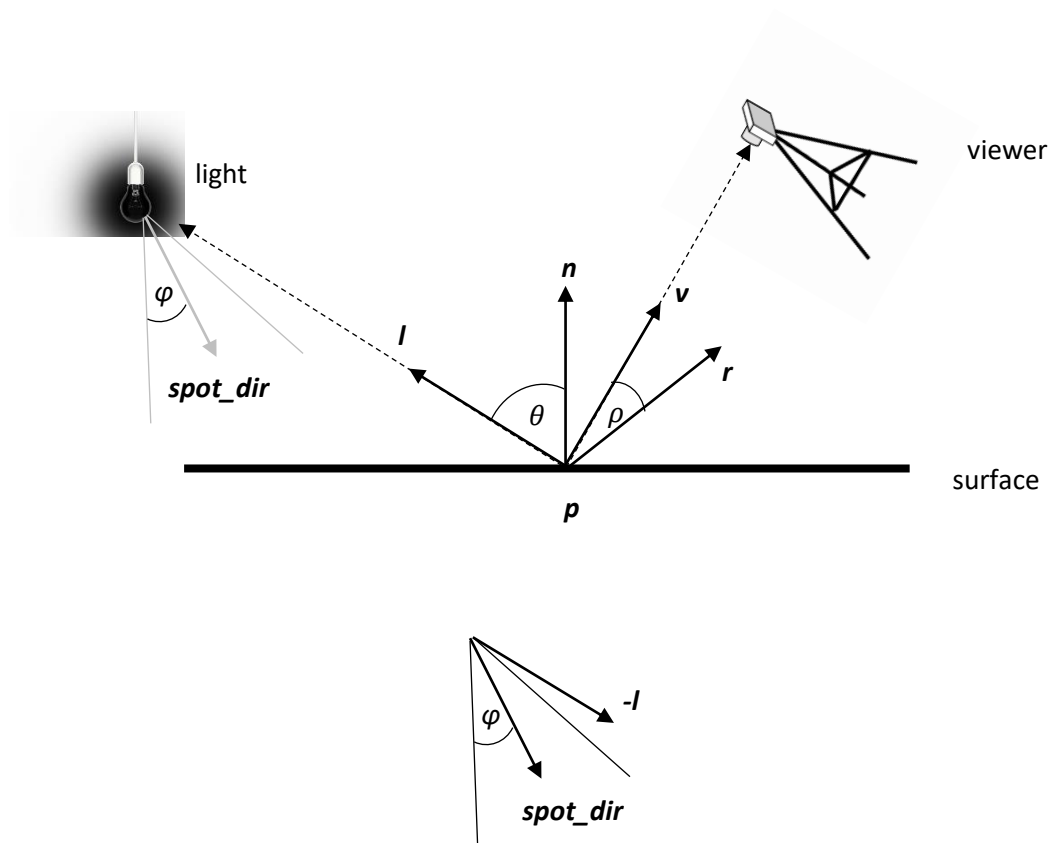
A *spot light* is a positional light which emits light in a cone shape and in a specified direction. Using a torch in a dark forest at night can be thought of as a spot light.

Any fragments that are inside of the cone are illuminated by the spot light, and any fragments that are outside of the cone are not illuminated by the spot light.



A spot light is defined by

1. a colour
2. a position, $light_pos$
3. a direction, $spot_dir$
4. a cutoff cone parameter, ϕ
5. constant attenuation, att_c
6. linear attenuation, att_l
7. quadratic attenuation, att_q



Then, check if the computed θ is smaller than the cut off angle φ of the cone of the spot light.

Phong's model can be used to calculate the total illumination of a fragment by a spot light in more or less exactly the same way as it was used to calculate the total illumination of a fragment by a positional light. The only difference is that Phong's model is made up of

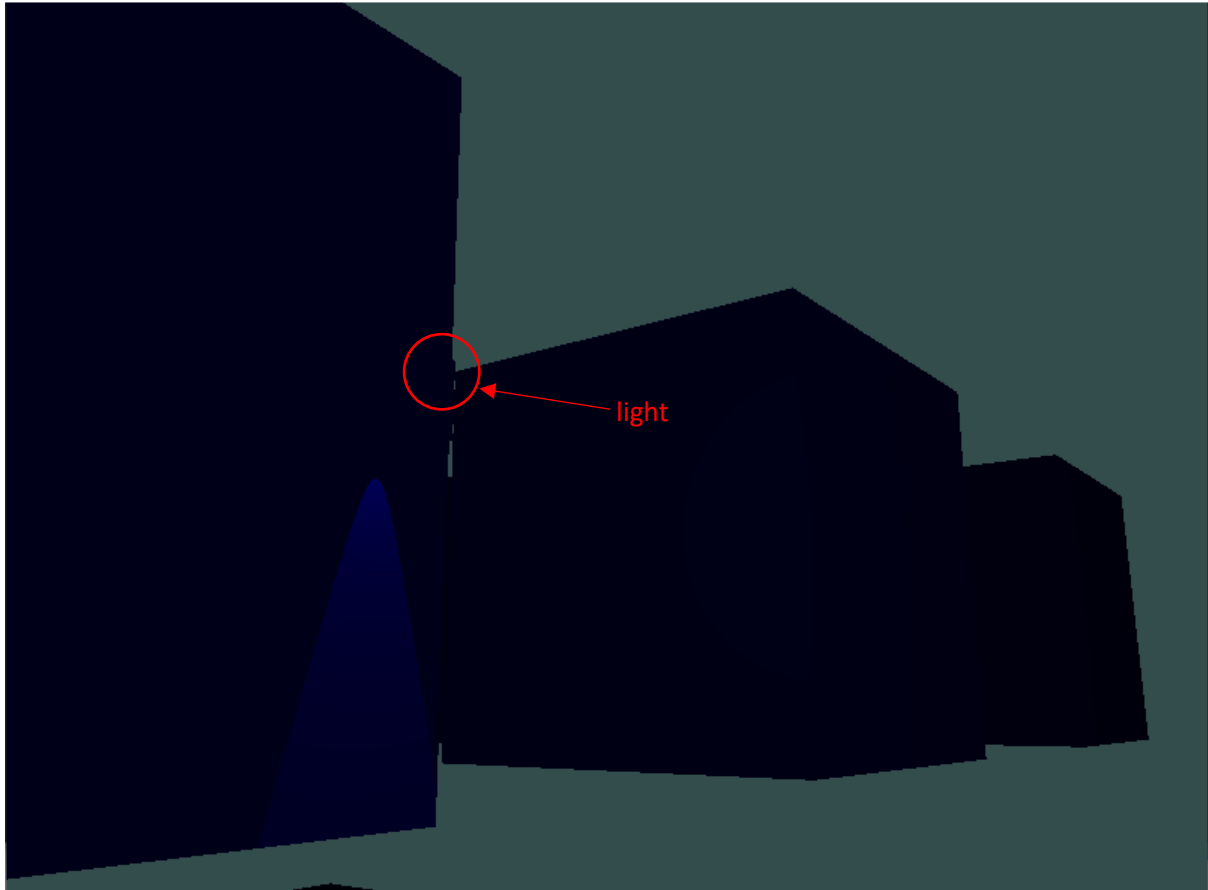
$$i_{phong} = (i_{amb} + i_{diff} + i_{spec}) * attenuation$$

if the fragment **is** lit by the spot light

$$i_{phong} = i_{amb} * attenuation$$

if the fragment **is not** lit by the spot light

In the image below, we can see that one surface has been illuminated by the cone of the spot light. The light itself is indicated with a red circle, and it's direction is pointing straight downwards.



Bibliography

Lambert, J. (1760). *Photometria, sive de mensura et gradibus luminis, colorum et umbrae*. Eberhard Klett.

Phong, B. (1975). Illumination for Computer Generated Pictures. *Communications of the ACM*, 18(6), 311-317.